Fall 2003

Problem 1 (Fa03, Fa23). Show that the initial value problem

$$f''(z) = zf(z) \qquad f(0) = 1 \qquad f'(0) = 1$$

has an unique entire solution in the complex plane.

Problem 2 (Fa03). List eight nonisomorphic groups of order 36.

Problem 3 (Fa03). Let A be a 2×2 matrix with complex entries. Prove that the series $I + A + A^2 + \cdots$ converges if and only if every eigenvalue of A has absolute value less than 1.

Problem 4 (Fa03). Give an example of a nonconstant irreducible polynomial $f(x)$ over $\mathbb{Q}$ with the property that $f(x)$ does not factor into linear factors over the field $\mathbb{K} = \mathbb{Q}[x]/\langle f(x)\rangle$.

Problem 5 (Fa03). Let $\mathcal{C}_{[0,1]}$ denote the space of continuous functions on $[0, 1]$. Define

$$d(f, g) = \int_0^1 \frac{|f(x) - g(x)|}{1 + |f(x) - g(x)|}\,\mathrm{d}x$$

1. Show that d is a metric on $\mathcal{C}_{[0,1]}$.
2. Show that $(\mathcal{C}_{[0,1]}, d)$ is not a complete metric space.

Problem 6 (Fa03). Let $A(m, n)$ be the $m \times n$ matrix with entries

$$a_{ij} = j^i \qquad \text{for } 0 \leqslant i \leqslant m-1,\ 0 \leqslant j \leqslant n-1$$

Regarding the entries of $A(m, n)$ as representing congruence classes module p, determine the rank of $A(m, n)$ over the finite field $\mathbb{F}_p$ for all $m, n \geqslant 1$ and all primes p.

Problem 7 (Fa03). Let $D = \{z \in \mathbb{C} \mid |z| \leqslant 1\} \setminus \{1, -1\}$. Find an explicit continuous function $f : D \to \mathbb{R}$ satisfying all the following conditions:

- f is harmonic on the interior of D (the open unit disc),
- $f(z) = 1$ when $|z| = 1$ and $\Im z > 0$,
- $f(z) = -1$ when $|z| = 1$ and $\Im z < 0$.

Problem 8 (Fa03). Let p be a prime. How many automorphisms does the group $\mathbb{Z}_{p^2} \times \mathbb{Z}_p$ have?

Problem 9 (Fa03). Let $f : [0,1] \to [0,1]$ be an increasing, not necessarily strictly increasing, function such that

$$f\left(\sum_{j=1}^{\infty} a_j 3^{-j}\right) = \sum_{j=1}^{\infty} \frac{a_j}{2} 2^{-j}$$

whenever the a_j are 0 or 2. Prove that there is a constant C_0 such that

$$|f(x) - f(y)| \leqslant C_0 |x - y|^{\log 2 / \log 3}$$

for all $x, y \in [0,1]$.

Problem 10 (Fa77, Su82, Fa97, Sp01, Fa03, Fa08, Sp11, Fa12, Sp18, Sp21). Evaluate $\displaystyle\int_{-\infty}^{\infty} \frac{\mathrm{d}x}{1+x^{2n}}$, where n is a positive integer.

Problem 11 (Fa03). Let $u_{m,n}$ be an array of numbers for $1 \leqslant m \leqslant N$ and $1 \leqslant n \leqslant N$. Suppose that $u_{m,n} = 0$ when m is 1 or N, or when n is 1 or N. Suppose also that

$$u_{m,n} = \frac{1}{4}\left(u_{m-1,n} + u_{m+1,n} + u_{m,n-1} + u_{m,n+1}\right)$$

whenever $1 < m < N$ and $1 < n < N$. Show that all the $u_{m,n}$ are zero.

Problem 12 (Fa03). Let A, B be $n \times n$ complex unitary matrices. Prove that $|\det(A+B)| \leqslant 2^n$.

Problem 13 (Fa03). Let L be a line in $\mathbb{C}$, and let f be an entire function such that $f(\mathbb{C}) \cap L = \emptyset$. Prove that f is constant.

Problem 14 (Fa03). Let n be a positive integer. Let $\varphi(n)$ be the Euler *totient* function, $\varphi(n) = \#\mathbb{Z}_n^*$. Prove that if $\gcd\{n, \varphi(n)\} > 1$, then there exists a noncyclic group of order n.

Problem 15 (Fa03). Let $f(z)$ be a meromorphic function on the complex plane. Suppose that for every polynomial $p(z) \in \mathbb{C}[z]$ and every closed contour Γ avoiding the poles of f, we have

$$\int_\Gamma p(z)^2 f(z)\,\mathrm{d}z = 0$$

Prove that $f(z)$ is entire.

Problem 16 (Fa03). Let G be a finite group and let X be the set of pairs of commuting elements of G:

$$X = \{(g,h) \in G \times G \mid gh = hg\}$$

1. Prove that $|X| = c|G|$ where c is the number of conjugacy classes in G.

2. Compute the number of pairs of commuting permutations on five letters.

Problem 17 (Fa03). The set of 5×5 complex matrices A satisfying $A^3 = A^2$ is a union of conjugacy classes. How many conjugacy classes?

Problem 18 (Fa03). Let $\lambda, a \in \mathbb{R}$, with $a < 0$. Let $u(x,y)$ be an infinitely differentiable function defined on an open neighborhood of the closed unit disc $\overline{\mathbb{D}}$ such that

$$\begin{cases} \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = \lambda u & \text{in } \mathbb{D} \\ D_n u = au & \text{in } \partial\overline{\mathbb{D}} \end{cases}$$

Here $D_n u$ denotes the directional derivative of u in the direction of the outward unit normal. Prove that if u is not identically zero in $\mathbb{D}$ then $\lambda < 0$.